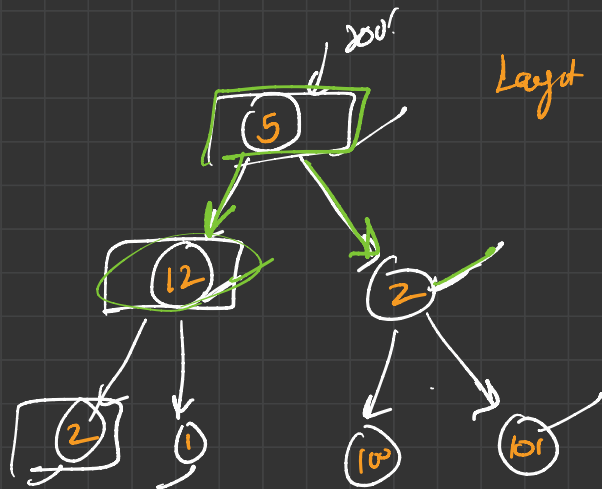



Greedy Algorithms

what $\rightarrow$?

local optimum

moment



Leaf $\rightarrow$ Root to
Leaf node

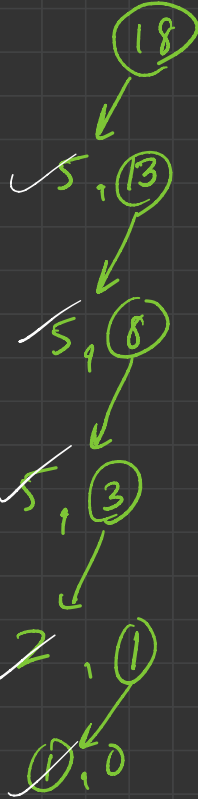
$$5 + 12 + 2 \rightarrow 19$$



$$5 + 2 + 101 \rightarrow 108$$

Target $\rightarrow 18 \neq$

5 coin $\times$
2 coin $\times$
1 coin



5 coins $\rightarrow 5 + 5 + 5 + 2 + 1$

$\rightarrow$ cost $\rightarrow$ custom up
 $\rightarrow$ max
 $\rightarrow$ min
 $\rightarrow$ pq
 $\rightarrow$ set

① N meetings in One room



N meetg

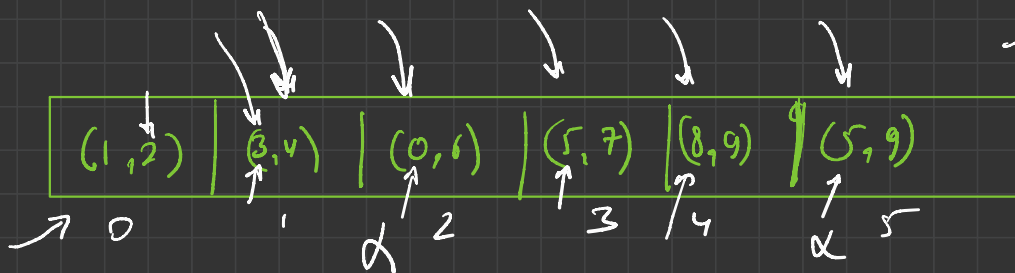
start → { 1, 3, 0, 5, 8, 5 }

end → { 2, 4, 6, 7, 9, 9 }

Green arrows point from the start and end arrays to the corresponding elements in the sorted array below.

int:-

count = 1



sort basis on
ending time

ansStart = 1
ansEnd = 2

$$3 > 2$$

$$\boxed{\text{count} = 2}$$

$$\text{ansEnd} = 3$$



$$0 > 3 \rightarrow \text{f}$$

$$8 > 3$$

$$\text{count} = 3$$

$$\text{ansEnd} = 5$$



$$8 > 5$$

$$\text{count} = 4$$

$$\text{ansEnd} = 8$$



$$5 > 8 \rightarrow \text{f}$$

(2) Shop in Candy Store

N - Candies

→ N=4

(K=2)

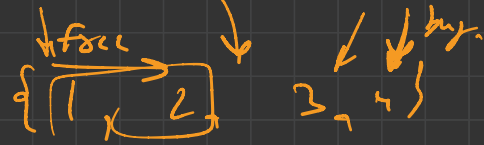
Candies [] = { 3, 2, 1, 4 }

min
amount
of money

N=4, K=2

C[] = { 3, 2, 1, 4 } → [1], 2, 3, 4

3 → [1] → [1, 3, 4]
2 → [1, 2, 3, 4]
4 → [4] → [1, 2, 4]
3 → [1, 2, 3, 4]



①

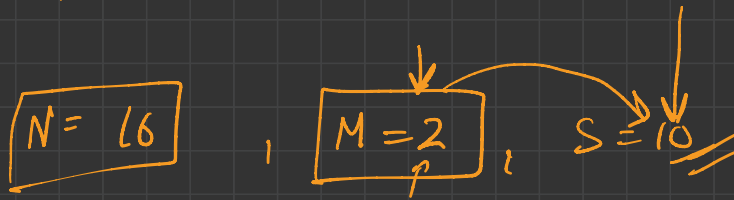
$i/p \rightarrow N \rightarrow \overset{\text{max}}{\text{food per day}}$

$\rightarrow S \rightarrow \text{days to survive}$

$\rightarrow M \rightarrow \text{food per pay to survive}$

Monday

Survive $\rightarrow$ done



$$\text{total food} = \boxed{S * M} = 20$$

1st day $\rightarrow \boxed{16 \text{ unit}}$ $\rightarrow$ 8 day

gth day $\rightarrow \boxed{16 \text{ unit}}$ $\rightarrow$ 8 day

$$\frac{20}{16} = \boxed{1} \text{ days} + 1$$

16 days $\boxed{4}$

$$\boxed{16} + \boxed{4}$$

1 day

②

$$N = 20$$

$$M = 30$$

$$S = 10$$

$$\text{if } (S * M) \% N == 0$$

$$\text{return } \frac{(S * M)}{N}$$

else

$$\text{return } \left(\frac{(S * M)}{N} \right) + 1$$

$$-1$$

$$N = 1$$

$$M = 1$$

$$S = 10$$

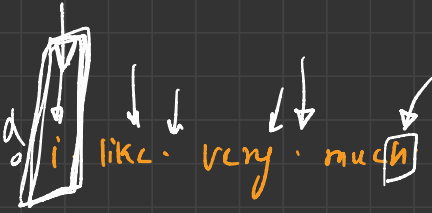
1 day → Sat

9 days

$$\text{ans} \rightarrow 10 \rightarrow \text{days}$$

M →
T
W
T
Fri
Sat
Sun X

①

S =  i like · very · much

o/p → much · very · like · i

h c u m
i → much

y r e v
i → very

(1)

M students

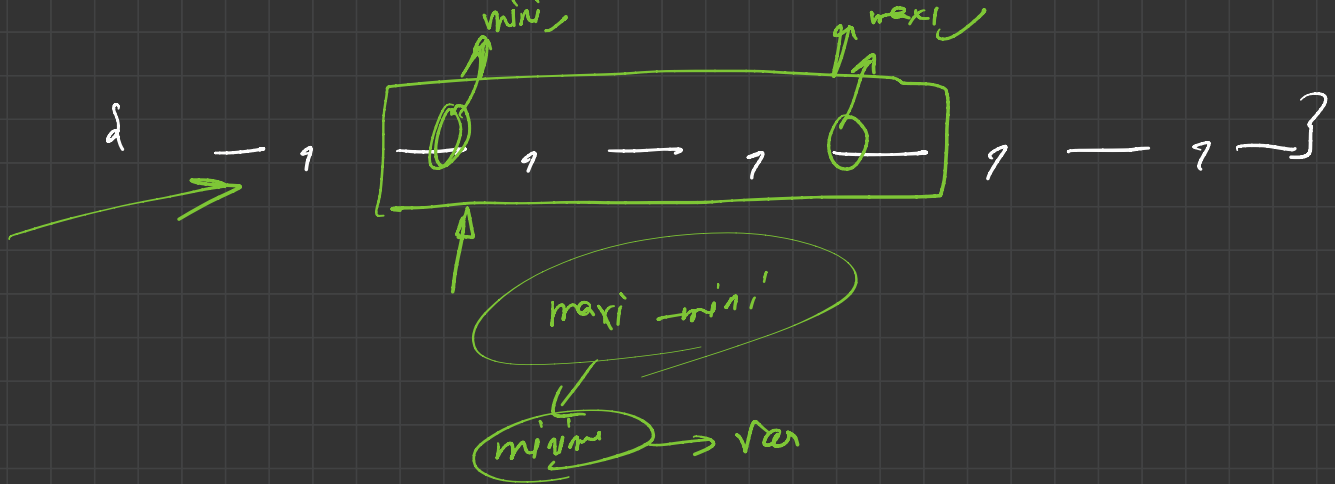


N packet
↓

$$M \leq N$$

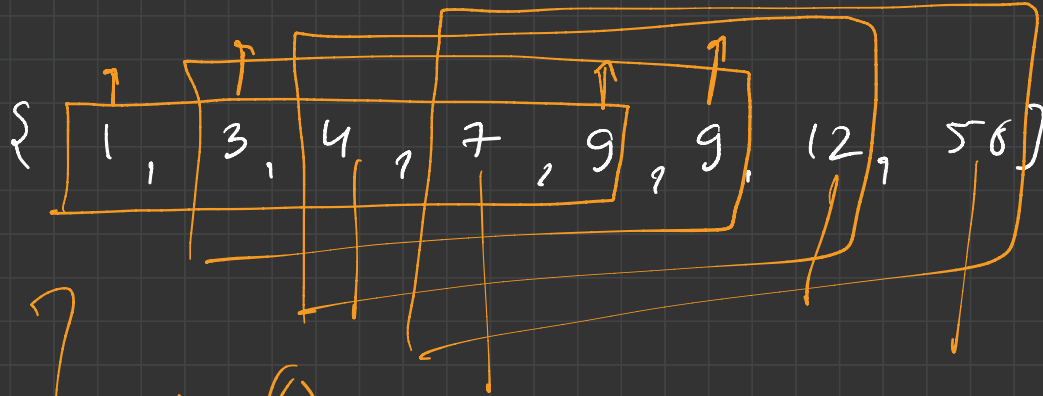
1 — 2 — 3 — 4 — 5 — }

$$M = 3$$



$$N = 8, \quad M = \underline{\underline{5}}$$

→ { 3, 4, 1, 9, 56, 7, 9, 12 }



$$\left. \begin{array}{l} 9-1=8 \\ 9-3=6 \\ 12-4 \rightarrow 8 \\ 56-7 \rightarrow 49 \end{array} \right\} \text{mi} \rightarrow 6$$

→ Min Cost of ropes

N ropes

↳

sort $\{4, 3, 2, 6\}$

cost = 5

sort $\{4, 5, 6\}$

cost = 9

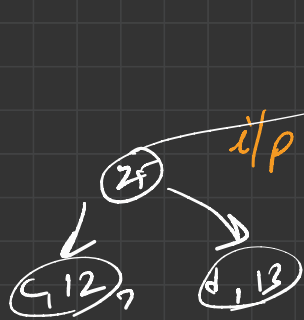
sort $\{9, 6\}$

cost = 15

29

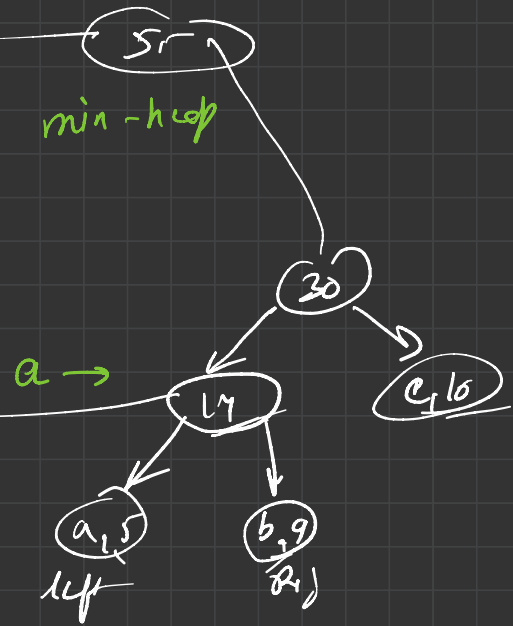
15

→ min. heap
 $O(1)$



$S = \text{"abcdef"}$

$freq \rightarrow [5, 9, 12, 13, 14, 45]$



Left → 0

Right → 1

11-2

f → 0

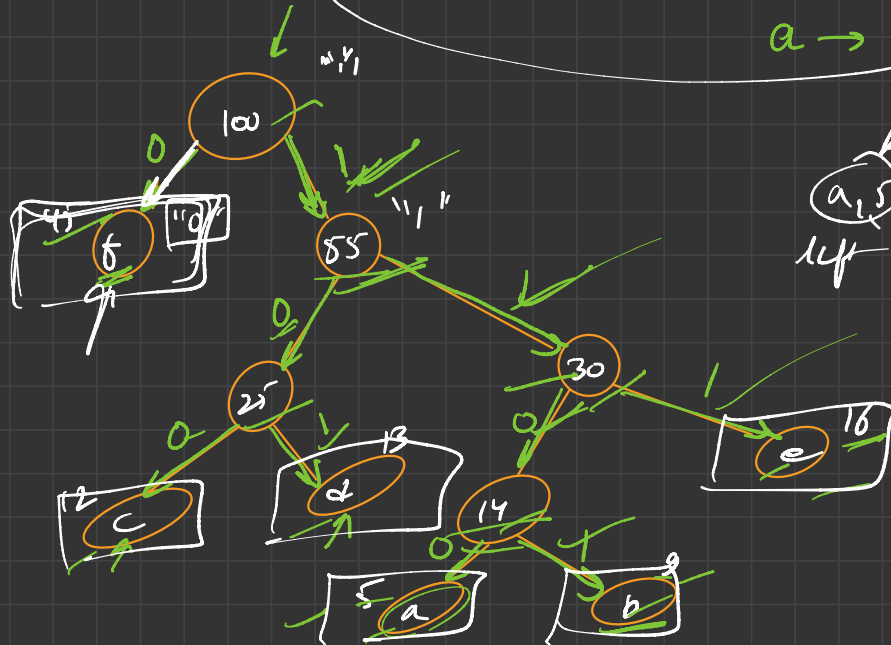
c → 100

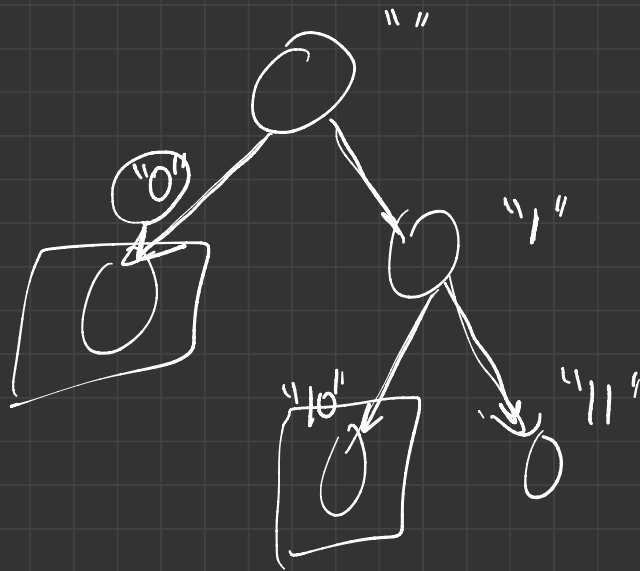
d → 101

a → 1100

b → 1101

e → 111





→ Fractional Knapsack:

N items

wt → {

val → [

N=3,

W=50

val →

wt →

60	100	120
10	20	30
1 st	2 nd	3 rd

W=50

10 kg
Ist
↓
60

20 kg
IInd
↓
100

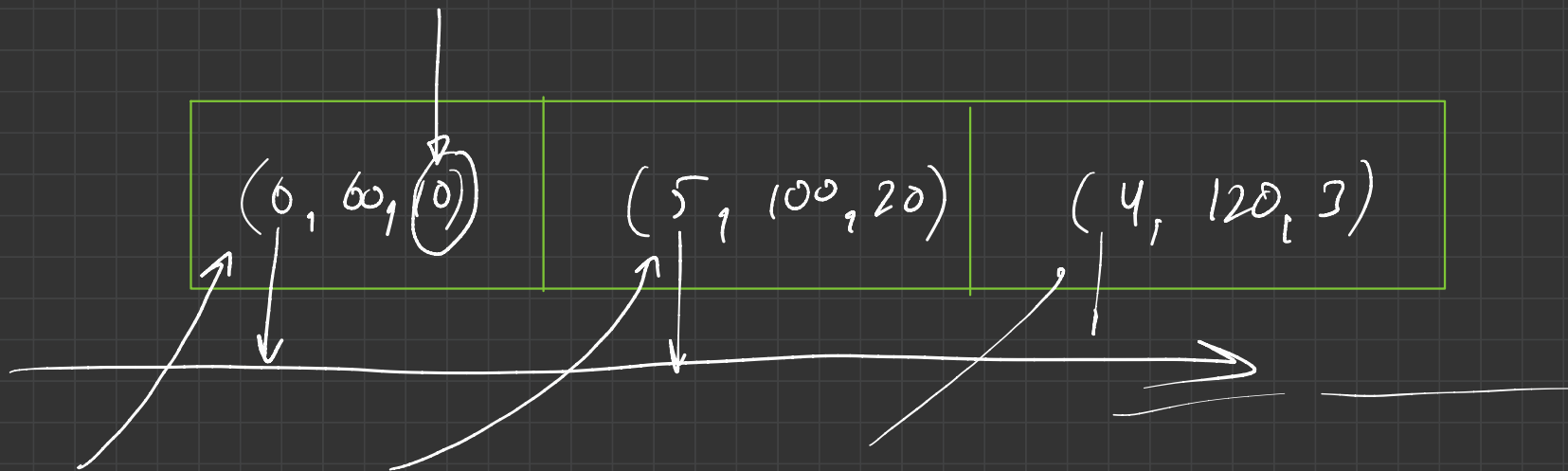
20 kg
IIIrd item
↓
20 × 4 = 80

240

$$I^{st} \rightarrow \frac{60}{10} = 6$$

$$II^{nd} \rightarrow \frac{100}{20} = 5$$

$$III \rightarrow \frac{120}{30} = 4$$



→ Job Sequencing Problem:-

i/p

Job Ids	Deadline	Profit
1	4	20
2	1	10
3	1	40
4	1	30

Handwritten table with 4 columns, each containing a job tuple (profit, id, deadline). Arrows indicate the current job (cur) and its index (cur_idx).

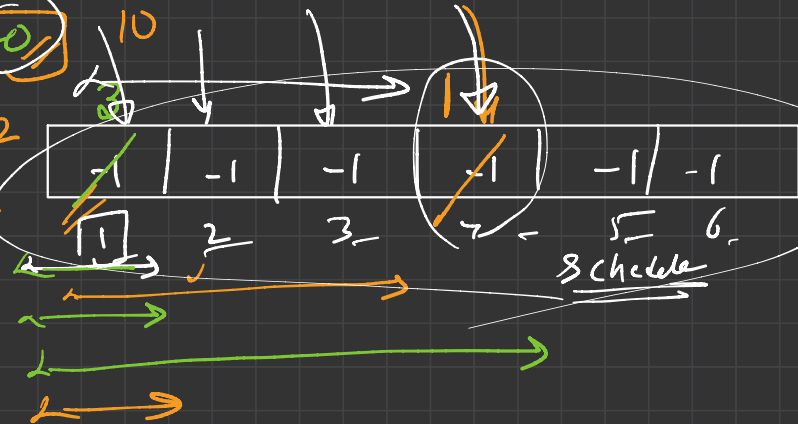
$(40, 3, 1)$ $\downarrow$ p $\downarrow$ id $\downarrow$ deadline cur	 $(30, 4, 1)$ cur_idx 	$(20, 1, 4)$ cur	 $(10, 2, 1)$ cur_idx
---	---	---------------------	---

cur Job Index $\rightarrow$ ~~0~~ ~~1~~ ~~2~~ 3 4

profit = ~~10~~ 30 ~~20~~ 10

id $\rightarrow$ 3 ~~4~~ ~~1~~ ~~2~~

deadline $\rightarrow$ ~~1~~ ~~4~~ ~~1~~ ~~2~~



29 60